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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Xeno Force
Selected Domain	Domain 1 — The Aiversecity Future-Ready Workforce Domain
Selected SDG Track (1–17)	8 *(SDG 8: Decent Work and Economic Growth)*
Project / Solution Name	ArcTalent
Name (Leader)	Emmanuel Heng Zi Teng
Email & Phone (Leader)	Enuelhzt1524@gmail.com 010-271-6363
University Affiliation(s)	Universiti Malaya
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Rapid artificial intelligence adoption is transforming workplace skill requirements faster than traditional curricula and institutional frameworks can adapt. In Malaysia, 56% of professionals already utilise AI tools at work, yet merely 26% have received structured formal training (Hays Asia Salary Guide 2025). Furthermore, MOSTI projects that over 600,000 Malaysian workers will require substantial reskilling within three to five years to avoid economic displacement.

This challenge acutely impacts working adults and Small-and-Medium Enterprises (SMEs), which form 97% of Malaysian businesses. Working adults cannot afford to quit employment or commit to multi-year degrees; they operate under stringent real-world constraints—often having fewer than four hours per week and zero discretionary training budget. Simultaneously, resource-constrained SMEs lack enterprise HR intelligence to forecast emerging skill gaps, leading to premature redundancies or costly external recruitment rather than internal redeployment.

The socioeconomic consequences are severe: widening wage inequality, heightened vulnerability for mid-career workers, and lost productivity. Environmentally and socially, as green and digital transformations converge, vulnerable workers risk exclusion from high-value future roles in clean technology, sustainable operations, and responsible AI governance. Without accessible, agile, and transparent reskilling mechanisms, workforce disruption threatens equitable economic development and social mobility.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

ArcTalent is an adaptive, explainable career reskilling command center that empowers working adults and SME managers to navigate changing labour markets across two distinct AI scenarios: **Moderate** and **Rapid AI Adoption**.

Rather than relying on static course directories or opaque "black-box" predictions, ArcTalent implements an explainable, multi-tiered AI architecture:

1. **Evidence-Grounded GenAI Skill Extraction**: Leveraging cloud-hosted GenAI APIs (Qwen/Qwen2.5-7B-Instruct on **RunPod** via Vercel AI SDK), ArcTalent processes synthetic CVs (PDF/text) to extract verified skill candidates anchored to exact verbatim textual evidence, preventing hallucinations and enforcing human-in-the-loop confirmation.

2. **Deterministic, Explainable Matching Engine**: Evaluates adjacent career suitability across four transparent pillars: current transferable skill coverage (55%), work-family interest (15%), market demand signals (15%), and scenario resilience (15%). Crucially, ArcTalent computes the **"shared foundation"**—skills vital across both moderate and rapid futures (e.g., data literacy and critical thinking)—enabling workers to build no-regret capabilities immediately.

3. **Constraint-Aware Greedy Scheduler**: Automatically packs verified, public zero-cost learning modules strictly within learner constraints (e.g., 4 hours/week, RM0 budget) while respecting course and skill prerequisites.

4. **SME Workforce Redeployment**: Provides managers with an 8-worker gap heatmap, internal capacity slot controls, grouped training economies, and UN SDG mappings (SDG 4.4, 8.2, 8.5, 10.2).

Incorporating SDC Phase 1 workshop learnings, we integrated sponsor cloud hosting (**Replit**) for our full-stack Next.js deployment, GPU cloud infrastructure (**RunPod**) for LLM inference, GenAI structured outputs for deterministic parsing, and strict TypeScript/Zod schemas to ensure enterprise-grade data integrity.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

1. Core Technologies, Frameworks & Libraries:

- **Frontend & Application Framework**: Next.js 16 (App Router), React 19, TypeScript, Lucide React, Custom CSS (Orion dark command center design system).
- **Validation & Logic Engine**: Zod for strict schema validation; deterministic heuristic matching algorithms and greedy heuristic scheduling (engine.ts).
- **Document Processing**: pdf-parse for client-uploaded synthetic text PDF parsing.
- **Client Storage & Privacy**: Browser LocalStorage for profiles, scenario settings, and course tracking, ensuring zero persistent storage of personal data on the server.

2. Official SDC Sponsor Cloud & GenAI Platforms Utilized:

- **Replit (Official SDC Sponsor)**: Serves as the primary production cloud platform, hosting the full-stack Next.js web application, route handlers (/api/analyze, /api/extract, /api/pdf, /api/signals, /api/health), and live deployment at https://arc-talent--emmanuelheng.replit.app/.
- **RunPod (Official SDC Sponsor)**: Hosts our dedicated cloud GPU instance running Qwen/Qwen2.5-7B-Instruct via OpenAI-compatible endpoints. The Vercel AI SDK connects to this endpoint using structured schema decoding to extract skill candidates with exact verbatim evidence quotes.
- **External Signal Adapters**: Arbeitnow Job Board API & arXiv cs.AI RSS feeds for real-time market demand context.

3. System Architecture Flow:

[Client: Orion UI / LocalStorage]

↕ HTTPS / JSON

[Replit Next.js App Server: App Router & API Endpoints]

├── /api/analyze ──> [Deterministic Matching Engine & Greedy Scheduler]

├── /api/pdf ──> [pdf-parse Stream Extraction]

├── /api/signals ──> [Arbeitnow & arXiv Public Feed Adapters]

└── /api/extract ──> [RunPod GPU Cloud: Qwen2.5-7B LLM via AI SDK]

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied design-thinking by centering on the lived realities of time-poor Malaysian workers and resource-constrained SME owners:

1. **Empathize & Define**: Mid-career workers expressed exhaustion with multi-year degrees and opaque AI scoring that felt dismissive of their background. SMEs needed immediate internal redeployment viability rather than complex enterprise software. We framed our core persona, "Aina" (4 hrs/week, RM0 budget, 8-week horizon).
2. **Ideate**: We reframed reskilling from "all-or-nothing retraining" to incremental, transferable "skill bridges" and shared foundations that minimize career risk under uncertainty.
3. **Prototype & Test**: Engineered the Orion command center featuring 3-question learning check-ins, interactive SVG constellation maps, and instant dated scenario replanning. Every workflow was validated against 32 automated unit tests and 14 production API integration checks.
4. **Phase 2 Deployment Viability**:
    - Architected a zero-database, privacy-first local storage model that eliminates enterprise data compliance hurdles.
    - Live on Replit cloud with RunPod GPU acceleration.
    - Phase 2 is structured for direct drop-in connection with official Malaysian HRD Corp claimable programmes and national job vacancy feeds (MYFutureJobs).

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary Targets:** • **SDG Target 4.4**: *"By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship."* • **SDG Target 8.2**: *"Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors."* • **SDG Target 8.5**: *"By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value."* **Supporting Target:** • **SDG Target 10.2**: *"By 2030, empower and promote the social, economic and political inclusion of all, irrespective of age, disability, race, ethnicity, origin, religion or economic or other status."*

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**74% of AI-active professionals in Malaysia lack formal training**: While 56% of Malaysian professionals actively use AI tools at work, only 26% have received structured formal training—leaving a **74% untrained deficit** *(Source: Hays Asia Salary Guide 2025)*. **>600,000 Malaysian workers face imminent disruption**: The Ministry of Science, Technology and Innovation (MOSTI) estimates that over **600,000 Malaysian workers** will require comprehensive reskilling within 3 to 5 years due to rapid AI automation *(Source: MOSTI Malaysia)*. **Severe Time & Budget Constraints**: Mid-career working adults have an average of **<4 discretionary study hours per week** and **RM0 personal training budget**, while 97% of Malaysian businesses (SMEs) lack HR analytics to identify internal redeployment opportunities before initiating redundancies or costly external hiring.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Skill Gap Pivot Distance Reduction (%)**: Tracking the mathematical reduction in skill deficit percentage, computed as (1 − skill_coverage) × 100 across 16 transferable work skills. **SME Training Cost & Hour Efficiency**: • Percentage (%) training expenditure savings achieved per employee through internal grouped training cohorts vs. individual commercial enrolments. • Total planned learning person-hours delivered within weekly working-adult constraints (≤4 hours/week).

Metric Category	Your Response
	□ **Dual-Scenario "No-Regret" Foundation Ratio (%)**: The proportion of early scheduled study hours mapped strictly to the "shared foundation" (skills required across both Moderate and Rapid AI adoption scenarios). □ **Skill Evidence Verification Rate (%)**: 100% of GenAI-extracted candidate skills backed by verbatim, inspectable CV text substrings with zero hallucination.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	□ **40% Reduction in Average Skill Gap Distance**: Synthetic and pilot workers achieve at least a **40% reduction in pivot distance** toward an adjacent high-resilience destination within an 8-week horizon under a RM0 budget constraint. □ **35% Training Cost Reduction for SMEs**: Achieving a **35% reduction in enterprise reskilling expenditure** through automated grouped course scheduling and internal opening redeployment across an 8-worker SME cohort. □ **100% Shared Foundation Coverage in Initial Phase**: Guaranteeing that **100% of the first 4 weeks of study** builds transferable foundation capabilities (e.g., data literacy, critical thinking) that remain valid regardless of whether AI adoption proceeds at a moderate (0.35) or rapid (0.75) pace. □ **100% Verified Human-in-the-Loop Adoption**: Achieving **100% compliance** with user-confirmed skill candidate adoption, ensuring zero unreviewed algorithmic skill assignments.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

Name	Type (code/dataset/model)	License
Next.js 16	Code (Framework)	MIT License
React 19 & React-DOM	Code (UI Library)	MIT License
TypeScript	Code (Language / Tooling)	Apache License 2.0
Zod	Code (Schema Validation)	MIT License
Lucide React	Code (Iconography)	ISC License
pdf-parse	Code (PDF Text Extraction)	MIT License
Vercel AI SDK (`ai` , `@ai-sdk/openai-compatible`)	Code (AI Model Adapter)	Apache License 2.0
Qwen/Qwen2.5-7B-Instruct	Pre-trained AI Model (Hosted on RunPod)	Apache License 2.0
US O*NET Database	Dataset / Occupational Framework	CC BY 4.0
Arbeitnow API	External Job Feed Data	Public API Terms
arXiv cs.AI RSS	Research Publication Feed	Open Access

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date] 25.10.2026